

Francisco Zorrilla

Computational biologist

Postdoc, ETH Zürich (2024 to 2026) · PhD, University of Cambridge · EMBL Heidelberg

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Italian citizen, Swiss B permit

SUMMARY

I build open-source modelling tools for microbial communities and apply them to real questions in food, health and the environment. I have worked in metagenomics and metabolic modelling for 7+ years, turning genomes and metagenomes into predictions of what each microbe in a community consumes and produces, together with wet-lab partners in academia and industry. At ETH Zürich I co-led a multi-institution NCCR Microbiomes flagship project and mentored three PhD students. I work in R and Python with reproducible pipelines (Snakemake, Conda, Slurm). Open to Senior or Lead roles in Switzerland from December 2026.

Key numbers: 1,100+ citations (h-index 9) | 4 Nature journal papers | 2,600+ metaGEM downloads | 6 labs coordinated

HOW I COULD HELP AN R&D TEAM

- Predict from genomes which nutrients the strains in a collection cannot make and which partners could supply them, and check those predictions against growth data, including where they fail.
- Model how the strains in a starter culture or consortium feed each other, alongside fermentation and omics data.
- Build a draft genome-scale metabolic model of a production organism from its genome.
- Annotate genes that sequence similarity misses, using protein language models and structure search (ProstT5, Foldseek), and feed the results into metabolic models.
- Build reproducible analysis pipelines that other groups install and run on their own data.
- Turn a broad research goal into a plan with clear modules, bring the right people in to lead them, and keep them on track.

EXPERIENCE

Postdoctoral Researcher · Co-lead, NCCR Microbiomes Work Package 5 Flagship Project

ETH Zürich, Sunagawa Lab, Zürich, Switzerland | Oct 2024 - Sep 2026

- **Planning:** designed the 4-year computational roadmap and its modules for the flagship (2024 to 2028; ETH Zürich, EPFL, UZH, UNIL and CHUV) and pitched it to the PIs.
- **Leadership:** coordinated the flagship with a PhD co-lead from EPFL on behalf of the work-package leaders. I co-organised its launch workshop (Bern, March 2025), recruited the co-leads of the four modules, ran about 20 coordination meetings and the progress reports, and co-led two modules until September 2026, with 50+ working sessions. I am co-first author on both module manuscripts (in preparation).
- **Module 4, structure-based metabolic models (with EPFL):** the module adds protein-structure predictions to genome-scale metabolic model reconstruction and benchmarks the result against existing methods and experimental data. I chose ProstT5 and Foldseek over AlphaFold so annotation scales to microbiome gene sets, scripted and deployed GPU-enabled annotation on ETH's Euler cluster (Snakemake, Slurm) with custom enzyme and transporter databases, and designed the comparison against sequence search (MMseqs2, DIAMOND) and experimental growth phenotypes.
- **Module 3, machine learning for community assembly (with UZH):** the module trains a neural network on a large collection of gut microbiome samples to predict which species belong in a given community; it learns from co-occurrence during training but needs only gene annotations at inference. Working with a Sunagawa lab colleague, who built the model, and members of the von Mering lab, I designed and ran its validation, with co-occurrence, gene-content and phylogenetic baselines and an audit of the public training metadata that cleaned the training set.
- **Mentoring:** mentored three PhD students and supervised a master's thesis (2025).

PhD researcher

University of Cambridge, MRC Toxicology Unit, Patil Lab, Cambridge, UK | Oct 2020 - Aug 2024

- **Open-source software:** built and released metaGEM (Nucleic Acids Research 2021, first author), a Snakemake workflow packaged on Bioconda, and ran it on HPC clusters to reconstruct 14,000+ genome-scale metabolic models from 483 metagenomes. Cited 150+ times and used by other groups in their own studies; six releases (2021 to 2023) and replies to 88 of the 90 GitHub issues users opened.
- **Soil cross-feeding** (Nature Microbiology 2026, co-first author): co-designed the genome sequencing and modelling, and carried out most of the genomic analysis and metabolic modelling of 62 strains, compared with the Kost lab's growth measurements.

- **Collaborations:** metabolic modelling for cheese flavour with Chr. Hansen (Nat. Commun. 2023); auxotrophy analysis for drug tolerance (Nat. Microbiol. 2022); metabolic models and metabolomics for *C. difficile* resistance (bioRxiv 2024); all metagenome assemblies and genome bins for a plastic-enzyme survey (mBio 2021).

Computational Biologist

EMBL Heidelberg, Patil Lab, Heidelberg, Germany | Aug 2019 - Jul 2020

- Started developing metaGEM, which became my PhD project.

EDUCATION

PhD, Biology

University of Cambridge, MRC Toxicology Unit, Patil Lab, Cambridge, UK | 2020 - 2024

- Thesis: Omics-driven and constraint-based modelling of microbial community metabolism.

MSc, Biotechnology

Chalmers University of Technology, Gothenburg, Sweden | 2017 - 2019

- iGEM 2018, Chalmers-Gothenburg team: gold medal, and nominee for Best Model in the graduate section for COM-dFBA, the team's community dynamic flux balance analysis framework, which I led.
- Elected treasurer of the Society for Biological Engineering students at Chalmers; managed events and budgets.

BSc, Biological Systems Engineering

University of California, Davis, USA | 2013 - 2017

- Senior design project (team of three): a low-cost Arduino system for remote sensing of plant CO₂ uptake; I programmed the Arduino and assembled the sensor electronics.

SKILLS

- **Bioinformatics:** metaGEM, Snakemake, fastp, MEGAHIT, CONCOCT, MaxBin2, MetaBAT2, metaWRAP, GTDB-Tk, MMseqs2, DIAMOND
- **Systems biology:** CarveMe, SMETANA, MEMOTE, COBRApy, ReFramed, RAVEN, FBA / FVA
- **Protein AI tools (pretrained models):** ProST5, Foldseek, AlphaFold2
- **Methods:** model validation and benchmarking (baselines, cross-validation, bootstrap), pangenomics, phylogenetic regression, metabolomics
- **Languages and infrastructure:** R, Python (NumPy, SciPy), MATLAB, Bash, Slurm HPC, GPU jobs on Slurm, Conda/Bioconda packaging, Git/GitHub, Claude Code (agentic coding)

PUBLICATIONS

- **Zorrilla F**, Buric F, Patil KR, Zelezniak A (2021). metaGEM: reconstruction of genome scale metabolic models directly from metagenomes. *Nucleic Acids Research* 49: e126. doi:10.1093/nar/gkab815
- Yousif G*, **Zorrilla F***, Dash S, ... Kost C (2026). Obligate cross-feeding of metabolites is common in soil microbial communities. *Nature Microbiology* 11: 2436–2447. *Co-first authors. doi:10.1038/s41564-026-02457-6
- Melkonian C, **Zorrilla F**, Kjærboelling I, ... Zeidan AA (2023). Microbial interactions shape cheese flavour formation. *Nature Communications* 14: 8348. doi:10.1038/s41467-023-41059-2
- Ambat A, van den Berg NI, **Zorrilla F**, ... Scaria J (2024). Emergent metabolic interactions in resistance to *Clostridioides difficile* invasion. *bioRxiv*. doi:10.1101/2024.08.29.610284
- **Zorrilla F**, Kerkhoven EJ (2022). Reconstruction of genome-scale metabolic model for *Hansenula polymorpha* using RAVEN. *Methods in Molecular Biology*, pp. 271–290. doi:10.1007/978-1-0716-2399-2_16
- Zrimec J, Kokina M, Jonasson S, **Zorrilla F**, Zelezniak A (2021). Plastic-degrading potential across the global microbiome correlates with recent pollution trends. *mBio* 12: e02155-21. doi:10.1128/mBio.02155-21

Also: drug tolerance (*Nature Microbiology* 2022), Enterobacteriaceae (*Nature Microbiology* 2025), FROG (*bioRxiv* 2024), CarveMe-GutMicrobes (*bioRxiv* 2026), two iGEM interlab studies (*Communications Biology* 2020, *PLOS ONE* 2021, consortium author) and my PhD thesis (Cambridge, 2024). Full list on Google Scholar.

TEACHING AND PRESENTATIONS

- **Teaching:** five courses: EMBO Practical Courses (Bangalore 2024; online 2022), SymbNET at EMBL-EBI (online, 2022), Part III Systems Biology at Cambridge (2024) and the Microbial Community Genomics block course at ETH Zürich (2025); 75+ participants.
- **Talks:** selected talk at the 8th COBRA Conference (Galway, 2022); invited talk at the S2M2 Summer School (online, 2022); posters at COBRA 2024 and 2021 and an EMBO Workshop (Heidelberg, 2022).

LANGUAGES AND INTERESTS

- **Languages:** English and Spanish (native level), Italian (B1), French (A1), German (A1).
- **Interests:** open-source software, low-cost lab hardware (microcontrollers, 3D printing), mycology, music, running.